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Thermodynamic assessment of the Sb–Sm and Sm–Sn systems

X.J. Liu [a,b], M.H. Chen [a], Y. Lu [a,b], Y.X. Huang [a,b], D. Wang [a], Q. He [a], D.L. Zhao [c], C.P. Wang [a,b,*]

[a] Department of Materials Science and Engineering, College of Materials, Xiamen University, Xiamen 361005, PR China
[b] Research Center of Materials Design and Applications, Xiamen University, Xiamen 361005, PR China
[c] Beijing General Iron and Steel Research Institute, Beijing 100081, China

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ABSTRACT

The thermodynamic assessment of the Sb–Sm and Sm–Sn binary systems have been carried out by using the CALPHAD (CALculation of PHAse Diagrams) method on the basis of the experimental data including phase equilibria and thermodynamic properties. The Gibbs free energies of the solution phases (Liquid, Bcc, Bct, Hcp and Rhombohedral phase) were described by the subregular solution model with the Relich–Kister polynomial, and those of the intermetallic compounds were described by the two-sublattice model. A consistent set of thermodynamic parameters has been derived for describing the Gibbs free energies of each phase in the Sb–Sm and Sm–Sn systems. A reasonable agreement was obtained between the calculated results and experimental data.

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1. Introduction

The rare earth (RE) elements have gained particular attention for their favorable magnetic, optical, electric, mechanical properties and the alloys formed with rare earth metals have always found application as high-performance functional materials, such as permanent magnets, semiconductors, lead-free solder alloys and hydrogen storage materials [1–6]. By adding a small amount of rare earth elements, the mechanical properties of Mg-based alloys can be substantially improved for microstructure development [7]. It is well known that more detailed knowledge about the phase equilibria and thermodynamic properties of rare earth alloy systems is very important to enhance the development of new rare earth material with superior properties. In order to construct a primary thermodynamic database of phase diagram in multi-component rare earth alloy systems, the thermodynamic description of each lower-order system which forms a part of this database is necessary to carry out. A series of the thermodynamic assessment of rare earth binary systems have already been performed by our research group [8–12].

The purpose of this work is to present the thermodynamic assessment of the Sb–Sm and Sm–Sn binary systems by means of the CALPHAD method, which is a powerful tool to save cost and time during the development of new materials [13]. The thermodynamic parameters of each phase in the Sb–Sm and Sm–Sn systems are optimized by fitting the experimental data on phase equilibria and thermodynamic properties.

2. Experiment information

2.1. Sb–Sm system

Regarding to the Sb–Sm system, the information on the structure properties of the individual compounds was firstly available in the literature [14–16]. Then Borzone et al. [17] systematically investigated the properties of Sb–Sm intermetallic compounds by metallographic, X-ray analysis and differential thermal analysis (DTA). In the work of Borzone et al., the Sb–Sm system was reported to include five intermetallic phases: Sb_2Sm, Sb_3Sm_5, Sb_3Sm_4, SbSm and $SbSm_2$, and no phase diagram was proposed, but a tentative one drawn on the base of analogy with the Nd–Sb system was sketched.

Subsequently, two experimental descriptions of the phase relations in the Sb–Sm system were reported by Sadigov et al. [18] and Abulkhaev [19], respectively. The complete phase diagram of the Sb–Sm system was firstly constructed using differential thermal analysis, X-ray diffraction and microstructural analysis by Sadigov et al. [18], who confirmed four intermetallic compounds: SbSm (congruent melting at 1922 °C), Sb_2Sm, Sb_3Sm_4 and Sb_3Sm_5 (peritectic formations at 1372 °C, 1792 °C and 1732 °C, respectively). The additional existence of $SbSm_2$ compound in Ref. [17] was not found by Sadigov et al. [18]. Two eutectic reactions: L (4 at% Sm)↔(Sb)+Sb_2Sm at 602 °C and L (90 at% Sm)↔Sb_3Sm_5+(βSm) at 1027 °C were determined and a polymorphic

* Corresponding author at: Department of Materials Science and Engineering, College of Materials, Xiamen University, Xiamen 361005, PR China.
Tel.: +86 592 2180606; fax: +86 592 2187966.
E-mail address: wangcp@xmu.edu.cn (C.P. Wang).

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transformation: (βSm)↔(αSm) occurred at 734 °C was also found in the Sb–Sm system.

In 1992, Abulkhaev [19] also experimentally reinvestigated the Sb–Sm phase diagram in the whole composition region, where SbSm melted congruently at 1875 °C and Sb_2Sm, Sb_3Sm_4 and Sb_3Sm_5 formed by peritectic reactions at 900 °C, 1680 °C and 1275 °C, respectively. Two eutectic reactions: L (0.2 at% Sm)↔(Sb)+Sb_2Sm at 625 °C and L (90 at% Sm)↔Sb_3Sm_5+(γSm) at 950 °C were also determined. The polymorphic transformation of the Sb_3Sm_5 phase which appeared at 1020 °C was indentified for the first time. By comparing the phase diagrams reported by Sadigov et al. [18] and Abulkhaev [19], it can be found that these two phase diagrams significantly differ with regard to invariant temperatures. According to the comment of Abulkhaev [19], the higher temperatures of invariant reactions obtained in Ref. [18] may be caused by the oxidation of the alloys during synthesis. The phase diagram of the Sb–Sm system was reviewed by Okamoto [20], as shown in Fig. 1.

The experimental thermodynamic properties of the Sb–Sm system were reported by several authors [17,21–23]. Lebedev et al. [21] employed EMF (Electromotive forces) technique to investigate the thermodynamic characteristic of Sm in liquid Sb at infinite dilution at 1000 K. Viksman and Gordienko [22] determined the enthalpy of formation of the SbSm compound by vapor pressure measurement. In addition, the enthalpies of formation of the intermetallic compounds in the Sb–Sm system were measured by Borzone et al. [17] using a different direct isoperibol calorimetry. Colinet and Pasturel [23] also predicted the enthalpy values of the Sb–Sm intermetallic compounds with various compositions by Miedema's model.

The Sb–Sm system has been thermodynamically optimized by Cacciamani et al. [24] without considering the phase diagram newly reported by Abulkhaev [19]. In the present work, the thermodynamic assessment of the Sb–Sm system is re-optimized based on the phase diagram from Abulkhaev [19] and the related thermodynamic information.

2.2. Sm–Sn system

Using differential thermal analysis, metallographic and X-ray method, the phase diagram of the Sn–Sm binary system was investigated by Percheron [25], where the following six intermetallic phases: Sm_5Sn_3 (congruent melting at 1505 °C), $SmSn_3$ (congruent melting at 1090 °C), Sm_4Sn_3, Sm_5Sn_4, $Sm_{11}Sn_{10}$ and

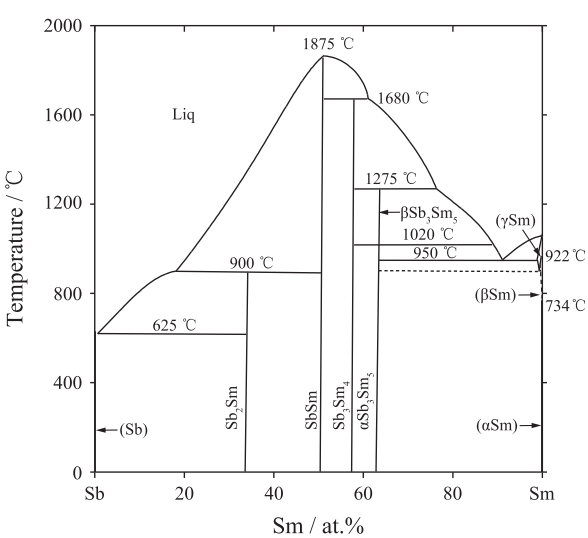

Fig. 1. The phase diagram of the Sb–Sm system reviewed by Okamoto [20].

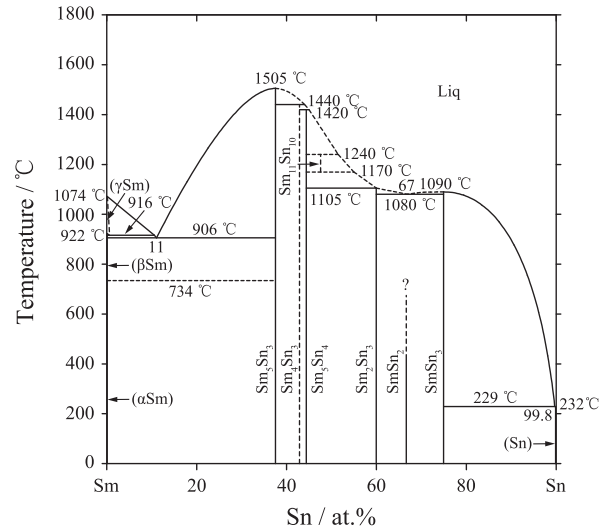

Fig. 2. The phase diagram of the Sm–Sn system drawn by Massalski [27].

Sm_2Sn_3 (peritectic formations at 1440 °C, 1420 °C, 1170 °C and 1105 °C, respectively) were verified. Percheron [25] also determined three eutectic reactions: L (11 at% Sn)↔(βSm)+Sm_5Sn_3 at 906 °C, L (67 at% Sn)↔Sm_2Sn_3+$SmSn_3$ at 1080 °C and L (99.8 at% Sn)↔$SmSn_3$+(Sn) at 229 °C and a polymorphic transformation: (βSm)↔(αSm) at 916 °C. But the trend of liquidus in the composition range between Sm_5Sn_3 and $SmSn_3$ compounds was not identified by Percheron [25].

Subsequently, Borzone et al. [26] briefly reviewed the phase diagram and thermodynamic information of the Sm–Sn binary system. In the work of Borzone et al. [26], the existence of $SmSn_2$ phase in the Sm–Sn system was firstly confirmed but the phase stability temperature of the $SmSn_2$ compound was not determined. During the thermodynamic optimization, the $SmSn_2$ phase as a pertectic compound was included in the Sm–Sn system. Fig. 2 shows the phase diagram of the Sm–Sn system compiled by Massalski et al. [27] according to the works mentioned above [25,26].

Peluso and Pool [28], Percheron et al. [29] and Meyer [30] measured the limiting partial enthalpies of Sm in liquid Sn at 750 K, 967.5 K and 1173 K, respectively by dissolution calorimetry at high dilution. The approximate range of the limiting partial enthalpy value at 828 K was determined by Pratt and Morris [31] using low solution rate calorimetry and a very negative value at 1000 K was obtained by Lebedev et al. [21] from EMF measurement. Yassin and Castanet [32] collected the partial enthalpy data of the Sm–Sn system from these literatures [21,28–31], and proposed a linear dependence on temperature for these selected data sets.

Additionally, Percheron et al. [29] measured the enthalpies of formation of the $SmSn_3$, Sm_2Sn_3 and Sm_5Sn_3 compounds at 295 K by liquid metal solution calorimetry. Using different techniques, the standard enthalpies of formation of the $SmSn_3$ and Sm_5Sn_3 phases were re-determined by Palenzona et al. [33] and Meschel and Kleppa [34], respectively. Colinet and Pasturel [35] predicted the enthalpies of formation of the intermetallic compounds in the Sm–Sn system with Miedma's model and Witusiewicz et al. [36] also obtained the evaluated enthalpy values based on the regularities of lanthanide compounds. The integral enthalpies of formation of the liquid Sm–Sn alloys at 1203 K were measured in the Sn-rich side ($0 < x(Sm) < 32$ at%) by Berrada et al. [37], who also estimated the limiting partial enthalpy of mixing of Sm in liquid Sn at 1203 K.

3. Thermodynamic models

The information about stable solid phases and the models used in the Sb–Sm and Sm–Sn binary systems is all listed in Table 1.

3.1. Solution phases

In the Sb–Sm and Sm–Sn binary systems (both A–B binary systems), the Gibbs free energies of the solution phases including Liquid, Bcc, Bct, Hcp and Rhombohedral phase were all described by the subregular solution model. The molar Gibbs free energy of each solution phase can be expressed as

$$G_m^{\phi} = {}^0G_A^{\phi}x_A + {}^0G_B^{\phi}x_B + RT(x_A \ln x_A + x_B \ln x_B) + {}^EG_m^{\phi} \quad (1)$$

where ${}^0G_i^{\phi}$ (i=A or B) is the molar Gibbs free energy of the pure component i in the respective reference state with the structure ϕ, which is taken from the SGTE pure element database compiled by Dinsdale [38]; x_i is the mole fraction of the component i; R is the gas constant; T is the absolute temperature; and the term ${}^EG_m^{\phi}$ is the molar excess Gibbs free energy, which is expressed in the form of Redlich–Kister polynomial [39] as

$${}^EG_m^{\phi} = x_A x_B \sum_{j=0}^{n} {}^jL_{A,B}^{\phi}(x_A - x_B)^j \quad (2)$$

where ${}^jL_{A,B}^{\phi}$ is the interaction energy between A and B atoms, and is described by the following expression:

$${}^jL_{A,B}^{\phi} = a + bT + cT^2 \quad (3)$$

where the parameters of a, b, and c are evaluated on the basis of experimental data in the present work.

3.2. Stoichiometric intermetallic compounds

The intermetallic compounds in the Sb–Sm and Sm–Sn binary systems were all treated as stoichiometric phases. The Gibbs free energy per mole of formula unit A_pB_q (p and q are the atomic numbers of A and B) can be expressed by the two-sublattice model, as follows:

$$G_{A:B}^{A_pB_q} = \Delta^0G_f^{A_pB_q} + \frac{p}{p+q}{}^0G_A^{SER} + \frac{q}{p+q}{}^0G_B^{SER} \quad (4)$$

where the $\Delta^0G_f^{A_pB_q}$ denotes the standard Gibbs free energy of formation per mole of formula unit A_pB_q from the pure elements, which is formulated as

$$\Delta^0G_f^{A_pB_q} = a' + b'T \quad (5)$$

where the parameters of a' and b' are optimized in the present work.

Table 1
The stable solid phases and the used models in the Sb–Sm and Sm–Sn systems.

System	Phase	Strukturbericht designation	Prototype	Modeling phase	Used models
Sb–Sm	(Sb)	*A*7	αAs	$(Sb, Sm)_1$	SSM
	Sb_2Sm	–	–	$(Sb)_{0.667}(Sm)_{0.333}$	SM
	SbSm	*B*1	NaCl	$(Sb)_{0.5}(Sm)_{0.5}$	SM
	Sb_3Sm_4	$D7_3$	Th_3P_4	$(Sb)_{0.429}(Sm)_{0.571}$	SM
	βSb_3Sm_5	–	–	$(Sb)_{0.375}(Sm)_{0.625}$	SM
	αSb_3SM_5	$D8_8$	Mn_5Si_3	$(Sb)_{0.375}(Sm)_{0.625}$	SM
	(γSm)	*A*2	W	$(Sb, Sm)_1$	SSM
	(βSm)	*A*3	Mg	$(Sb, Sm)_1$	SSM
	(αSm)	*C*19	αSm	$(Sb, Sm)_1$	SSM
Sm–Sn	(γSm)	*A*2	W	$(Sm, Sn)_1$	SSM
	(βSm)	*A*3	Mg	$(Sm, Sn)_1$	SSM
	(αSm)	*C*19	αSm	$(Sm, Sn)_1$	SSM
	Sm_5Sn_3	$D8_8$	Mn_5Si_3	$(Sm)_{0.625}(Sn)_{0.375}$	SM
	Sm_4Sn_3	$D7_3$	Th_3P_4	$(Sm)_{0.571}(Sn)_{0.429}$	SM
	Sm_5Sn_4	–	–	$(Sm)_{0.556}(Sn)_{0.444}$	SM
	$Sm_{11}Sn_{10}$	–	–	$(Sm)_{0.524}(Sn)_{0.476}$	SM
	Sm_2Sn_3	–	–	$(Sm)_{0.4}(Sn)_{0.6}$	SM
	$SmSn_3$	$L1_2$	$AuCu_3$	$(Sm)_{0.25}(Sn)_{0.75}$	SM
	(Sn)	*A*5	βSn	$(Sm, Sn)_1$	SSM

SM: sublattice model.
SSM: substitution solution model.

4. Optimized results and discussion

The optimization of the thermodynamic parameters of each phase in the Sb–Sm and Sm–Sn binary systems was carried out using the PARROT module in the Thermo-Calc software package [40], which works by minimizing the square sum of the differences between experimental data and calculated values. Various kinds of experimental data including phase diagram and thermodynamic properties were used as input to this program for optimization. During the assessment process, the selected information was given a certain weight according to the importance of data and the weight was changed by trial and error until most of the selected experimental data were reproduced within the expected uncertainty limits. In the present work, the literature data mentioned above have been selected for optimization. The thermodynamic parameters of each phase in the Sb–Sm and Sm–Sn systems were eventually optimized to obtain the best consistency between the calculated results and literature data, which are listed in Tables 2 and 3, respectively.

4.1. Sb–Sm system

The calculated phase diagram of the Sb–Sm binary system compared with experimental data [18,19] is shown in Fig. 3. A excellent agreement was obtained between the calculated results and the experimental phase diagram newly reported by Abulkhaev [19]. The calculated liquidus of the Sb-rich side slightly differ from the diagram of Abulkhaev, which is still needed to confirm by future experiments. All invariant reactions in the Sb–Sm system compared with the experimental data [18,19] are listed in Table 4. The calculated compositions and temperatures of all invariant reactions are in good accordance with experimental data [19].

As for the thermodynamic properties, the calculated standard enthalpies of formation of the intermetallic compounds with reference states of Sb (Rhombohedral) and Sm (Rhombohedral) phases at 298 K along with the literature data [17,22,23] are presented in Fig. 4. It can be seen that the present calculated results are in good consistence with the experimental data from Ref. [17]. The standard enthalpy of formation of the SbSm compound measured by Viksman and Gordienko [22] may be more negative than the calculated value while the enthalpy values predicted by Colinet and Pasturel [23] with Miedema's model seem to be large positive numbers.

4.2. Sm–Sn system

The calculated phase diagram of the Sm–Sn binary system together with the experimental data [27] is shown in Fig. 5. All invariant equilibria in the Sm–Sn system are summarized in Table 5, in which the experimental data [27] are also listed for comparison. It can be seen that most of the reported experimental data of phase diagram are well reproduced. The trend of the

Table 2
The optimized thermodynamic parameters of the Sb–Sm system.

Liquid phase, format $(Sb, Sm)_1$
${}^0L^{Liq}_{Sb,Sm} = -250,400-80T$
${}^1L^{Liq}_{Sb,Sm} = 2200+12T$
${}^2L^{Liq}_{Sb,Sm} = 44,500+43T$
${}^3L^{Liq}_{Sb,Sm} = -15,500$
Bcc phase, format $(Sb, Sm)_1$
${}^0L^{Bcc}_{Sb,Sm} = -257,800$
Hcp phase, format $(Sb, Sm)_1$
${}^0L^{Hcp}_{Sb,Sm} = 0$
Rhombohedral_A7 phase, format $(Sb, Sm)_1$
${}^0L^{Rhombohedral_A7}_{Sb,Sm} = 0$
Rhombohedral_Sm phase, format $(Sb, Sm)_1$
${}^0L^{Rhombohedral_Sm}_{Sb,Sm} = 0$
Sb_2Sm phase, format $(Sb)_{0.667}(Sm)_{0.333}$
${}^0G^{Sb_2Sm}_{Sb:Sm} = -89,500-0.16T+0.667{}^0G^{Rhombohedral}_{Sb}+0.333{}^0G^{Rhombohedral}_{Sm}$
SbSm phase, format $(Sb)_{0.5}(Sm)_{0.5}$
${}^0G^{SbSm}_{Sb:Sm} = -122000-6.21T+0.5{}^0G^{Rhombohedral}_{Sb}+0.5{}^0G^{Rhombohedral}_{Sm}$
Sb_3Sm_4 phase, format $(Sb)_{0.429}(Sm)_{0.571}$
${}^0G^{Sb_3Sm_4}_{Sb:Sm} = -118000-3.56T+0.429{}^0G^{Rhombohedral}_{Sb}+0.571{}^0G^{Rhombohedral}_{Sm}$
βSb_3Sm_5 phase, format $(Sb)_{0.375}(Sm)_{0.625}$
${}^0G^{Sb_3Sm_5}_{Sb:Sm} = -112012+1.25T+0.375{}^0G^{Rhombohedral}_{Sb}+0.625{}^0G^{Rhombohedral}_{Sm}$
αSb_3Sm_5 phase, format $(Sb)_{0.375}(Sm)_{0.625}$
${}^0G^{Sb_3Sm_5}_{Sb:Sm} = -112207+1.40T+0.375{}^0G^{Rhombohedral}_{Sb}+0.625{}^0G^{Rhombohedral}_{Sm}$

Table 3
The optimized thermodynamic parameters of the Sm–Sn system.

Liquid phase, format $(Sm, Sn)_1$
${}^0L^{Liq}_{Sm,Sb} = -225,800+95T-0.007T^2$
${}^1L^{Liq}_{Sm,Sn} = 3900-2T$
${}^2L^{Liq}_{Sm,Sn} = 16,800+5T$
${}^3L^{Liq}_{Sm,Sn} = 13,000$
Bcc phase, format $(Sm, Sn)_1$
${}^0L^{Bcc}_{Sm,Sn} = -56200$
Bct phase, format $(Sm, Sn)_1$
${}^0L^{Bct}_{Sm,Sn} = 0$
Hcp phase, format $(Sm, Sn)_1$
${}^0L^{Hcp}_{Sm,Sn} = 0$
Rhombohedral phase, format $(Sm, Sn)_1$
${}^0L^{Rhombohedral}_{Sm,Sn} = 0$
Sm_5Sn_3 phase, format $(Sm)_{0.625}(Sn)_{0.375}$
${}^0G^{Sm_5Sn_3}_{Sm:Sn} = -63,600+14.93T+0.625{}^0G^{Rhombohedral}_{Sm}+0.375{}^0G^{Bct}_{Sn}$
Sm_4Sn_3 phase, format $(Sm)_{0.571}(Sn)_{0.429}$
${}^0G^{Sm_4Sn_3}_{Sm:Sn} = -73,400+19.86T+0.571{}^0G^{Rhombohedral}_{Sm}+0.429{}^0G^{Bct}_{Sn}$
Sm_5Sn_4 phase, format $(Sm)_{0.556}(Sn)_{0.444}$
${}^0G^{Sm_5Sn_4}_{Sm:Sn} = -76,000+21.25T+0.556{}^0G^{Rhombohedral}_{Sm}+0.444{}^0G^{Bct}_{Sn}$
$Sm_{11}Sn_{10}$ phase, format $(Sm)_{0.524}(Sn)_{0.476}$
${}^0G^{Sm_{11}Sn_{10}}_{Sm:Sn} = -69,716+17.49T+0.524{}^0G^{Rhombohedral}_{Sm}+0.476{}^0G^{Bct}_{Sn}$
Sm_2Sn_3 phase, format $(Sm)_{0.4}(Sn)_{0.6}$
${}^0G^{Sm_2Sn_3}_{Sm:Sn} = -72,708+22.67T+0.4{}^0G^{Rhombohedral}_{Sm}+0.6{}^0G^{Bct}_{Sn}$
$SmSn_2$ phase, format $(Sm)_{0.333}(Sn)_{0.667}$
${}^0G^{SmSn_2}_{Sm:Sn} = -65,500+19.25T+0.333{}^0G^{Rhombohedral}_{Sm}+0.667{}^0G^{Bct}_{Sn}$
$SmSn_3$ phase, format $(Sm)_{0.25}(Sn)_{0.75}$
${}^0G^{SmSn_3}_{Sm:Sn} = -56,300+15.47T+0.25{}^0G^{Rhombohedral}_{Sm}+0.75{}^0G^{Bct}_{Sn}$

liquidus boundaries in the composition range between Sm_5Sn_3 and $SmSn_3$ compounds deviates from the dotted curves in Fig. 2, which has not been experimentally determined. The calculated compositions and temperatures for the eutectic reactions in the Sm–Sn system are in good agreement with the experimental results except for one eutectic reaction: $L \leftrightarrow SmSn_2 + SmSn_3$. The composition of this eutectic point at 1079.9 °C is calculated to be 71.2 at% Sn, which indicates a little higher than that reported in the literature (67 at% Sn). The pertectic temperature for the formation of the $SmSn_2$ phase is evaluated to be 1093.5 °C in the present work. These calculated results remain to be verified by more experiments in future.

The calculated standard enthalpies of formation of the intermetallic compounds with reference states of Sm (Rhombohedral) and Sn (Bct) phases at 295 K compared with the literature data [29,33–36] are presented in Fig. 6. It can be found that a satisfactory agreement is obtained between the calculated results and the selected experimental data in Refs. [29]. However, the enthalpies of formation of the $SmSn_3$ and Sm_5Sn_3 compounds measured by Palenzona et al. [33] and Meschel and Kleppa [34], respectively are probably too negative. As shown in Fig. 6, our calculated enthalpies of the Sm–Sn system show the same tendency with the data predicted by Colinet and Pasturel [35] and Witusiewicz et al. [36]. The calculated integral enthalpy of mixing of the liquid Sm–Sn alloys at 1203 K is consistent with the experimental result from Ref [37], as plotted in Fig. 7.

Fig. 8 is the calculated limiting partial enthalpy of Sm in liquid Sn versus temperature variation compared with the experimental data [21,28–31,37]. The calculated results are fitted linearly with the data sets determined by Peluso and Pool [28], Percheron et al.

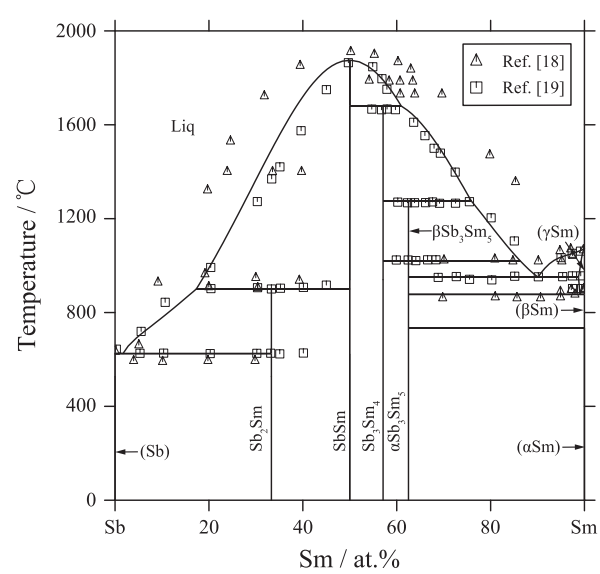

Fig. 3. The calculated phase diagram of the Sb–Sm system compared with the experimental data [18,19].

Table 4
Experimental and calculated invariant reactions in the Sb–Sm system.

Reactions	Temperature (°C)	Composition at% Sm			Type of reactions	References
$L \leftrightarrow (Sb) + Sb_2Sm$	602	4	~0	33.3	Eutectic	[18]
	625	0.2	0	33.3		[19]
	625.0	1.7	0	33.3		This work
$L + SbSm \leftrightarrow Sb_2Sm$	1372	21	50	33.3	Peritectic	[18]
	900	17.3	50	33.3		[19]
	900.1	17.3	50	33.3		This work
$L \leftrightarrow SbSm$	1922		50		Congruent	[18]
	1875		50			[19]
	1875.0		50			This work
$L + SbSm \leftrightarrow Sb_3Sm_4$	1792	66	50	57.1	Peritectic	[18]
	1680	59.7	50	57.1		[19]
	1680.2	60.9	50	57.1		This work
$L + Sb_3Sm_4 \leftrightarrow \beta Sb_3Sm_5$	1732	69	57.1	62.5	Peritectic	[18]
	1275	75.4	57.1	62.5		[19]
	1275.0	75.9	57.1	62.5		This work
$Sb_3Sm_4 + \beta Sb_3Sm_5 \leftrightarrow \alpha Sb_3Sm_5$	1020	88.1	57.1	62.5	polymorphic	[19]
	1020.2	86.3	57.1	62.5		This work
$L \leftrightarrow Sb_3Sm_5 + (\gamma Sm)$	1027	90	62.5	95	Eutectic	[18]
	950	90.0	62.5	99.1		[19]
	950.2	90.1	62.5	98.7		This work
$(\gamma Sm) \leftrightarrow Sb_3Sm_5 + (\beta Sm)$	877	92	62.5	95	Eutectoid	[18]
	–	–	62.5	~100		[19]
	877.0	99.0	62.5	100		This work

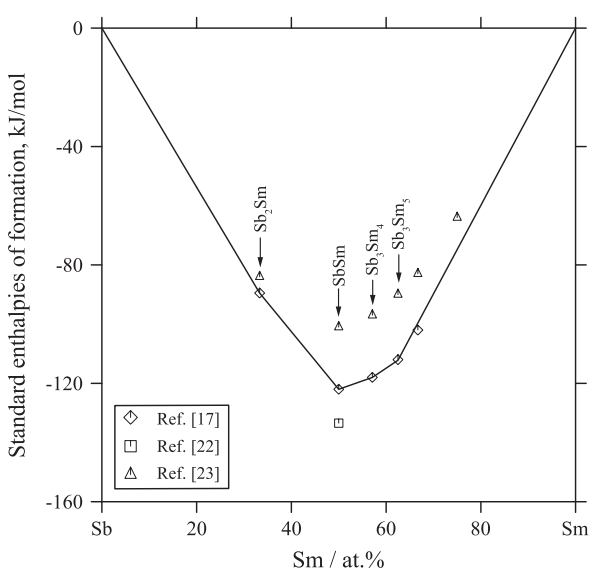

Fig. 4. The calculated enthalpies of formation of the intermetallic compounds with reference states of Sb (Rhombohedral) and Sm (Rhombohedral) phases at 298 K in the Sb–Sm system compared with the literature data [17,22,23].

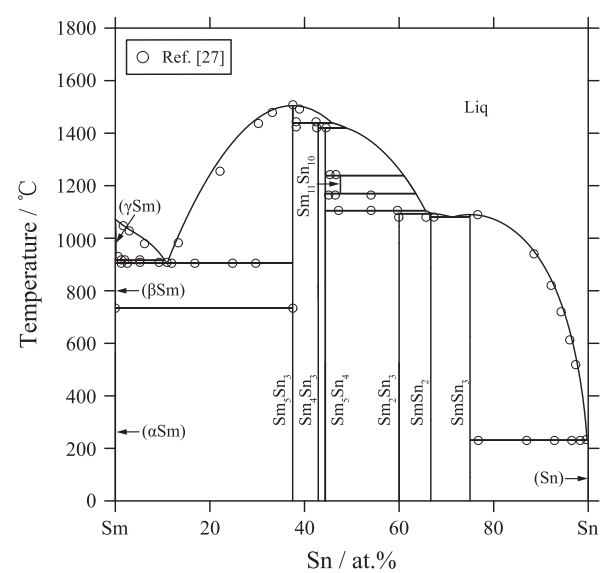

Fig. 5. The calculated phase diagram of the Sn–Sm system compared with the experimental data [27].

[29], Meyer [30] and Pratt and Morris [31], respectively using calorimetry, and two values obtained by Lebedev et al. [21] with EMF measurement and by Berrada et al. [37] with extrapolation from the corresponding integral enthalpy of mixing seem to be relatively too negative, which should not be taken into consideration in the present work. The similar conclusion has also been reached by Yassin and Castanet [32]. According to the literature [28,31,37], such extremely negative limiting partial enthalpies of

Table 5
Experimental and calculated invariant reactions in the Sm–Sn system.

Reactions	Temperature (°C)	Composition at% Sn			Type of reactions	References
(γSm)↔L+(βSm)	916	~0.1	10.5	0	Metatectic	[27]
	916.0	0.1	10.5	0		This work
L↔(βSm)+Sm_5Sn_3	906	11	0	37.5	Eutectic	[27]
	906.0	11.0	0	37.5		This work
L↔Sm_5Sn_3	1505		37.5		Congruent	[27]
	1505.1		37.5			This work
L+Sm_5Sn_3↔Sm_4Sn_3	1440	–	37.5	42.9	Peritectic	[27]
	1439.8	45.8	37.5	42.9		This work
L+Sm_4Sn_3↔ Sm_5Sn_4	1420	–	42.9	44.4	Peritectic	[27]
	1419.7	48.8	42.9	44.4		This work
L+Sm_5Sn_4↔$Sm_{11}Sn_{10}$	1240	–	44.4	47.6	Peritectic	[27]
	1238.6	61.0	44.4	47.6		This work
$Sm_{11}Sn_{10}$↔L+Sm_5Sn_4	1170	47.6	–	44.4	Metatectic	[27]
	1170.0	47.6	63.5	44.4		This work
L+Sm_5Sn_4↔Sm_2Sn_3	1105	–	44.4	60	Peritectic	[27]
	1105.0	65.6	44.4	60		This work
L+Sm_2Sn_3↔$SmSn_2$	1093.5	67.0	60	66.7	Pertectic	This work
L↔$SmSn_2$+$SmSn_3$	1080	66.7	67	75	Eutectic	[27]
	1079.9	66.7	71.2	75		This work
L↔$SmSn_3$	1090		75		Congruent	[27]
	1090.2		75			This work
L↔$SmSn_3$+(Sn)	229	99.8	75	~100	Eutectic	[27]
	231.0	99.7	75	100		This work

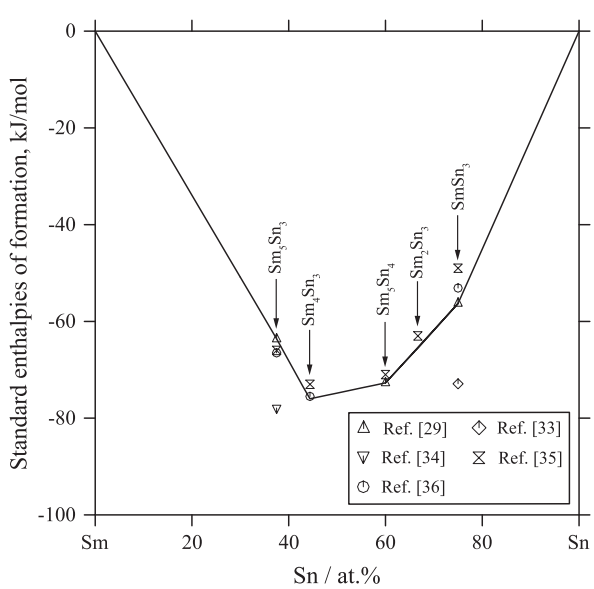

Fig. 6. The calculated enthalpies of formation of the intermetallic compounds with reference states of Sm (Rhombohedral) and Sn (Bct) phases at 295 K in the Sm–Sn system compared with the literature data [29,33–36].

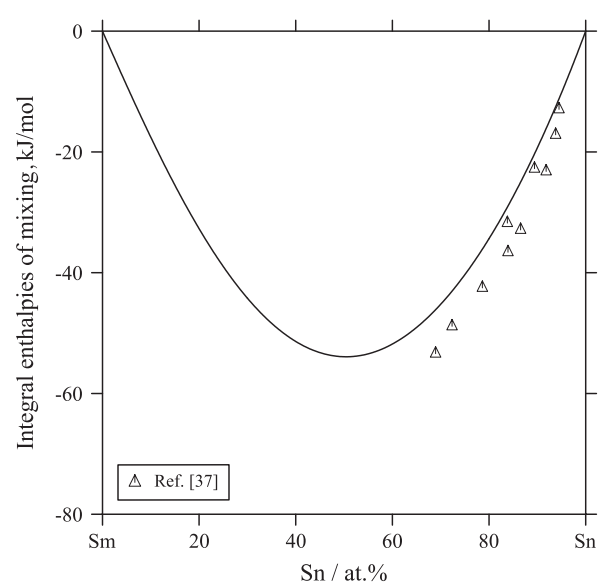

Fig. 7. The calculated integral enthalpy of mixing of the liquid Sm–Sn alloys at 1203 K compared with the experimental data [37].

the Sm–Sn solution were associated with strong short-range order in the liquid state. As a result, the inverted miscibility gap of liquid phase tends to be formed at high temperature. In order to avoid the formation of an unwanted inverted miscibility gap in the high-temperature liquid phase, a T^2 term of the excess parameter ${}^0L^L_{Sm,Sn}$ can be considered during the optimization of the Sm–Sn system with Redlich–Kister formalism.

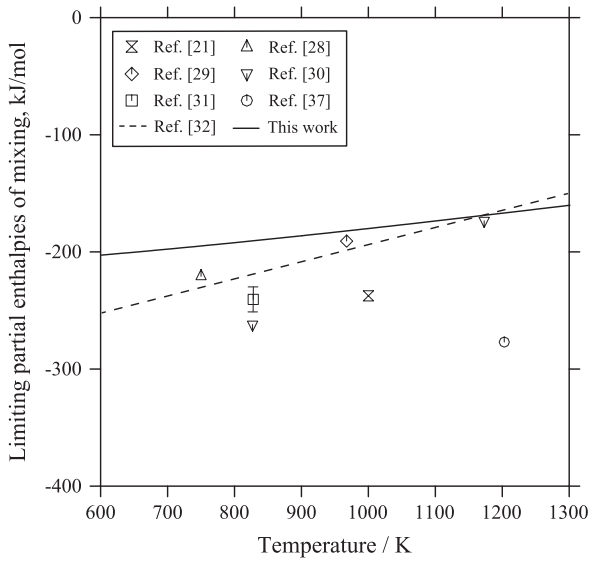

Fig. 8. The calculated limiting partial enthalpy of Sm in liquid Sn versus temperature variation compared with experimental data [21,28–32,37].

5. Conclusion

The phase diagram and thermodynamic properties in the Sb–Sm and Sm–Sn binary systems were evaluated by combining the thermodynamic models with experimental information. A consistent set of optimized thermodynamic parameters has been derived to describe the Gibbs free energy of each solution phase and intermetallic compound in the Sb–Sm and Sm–Sn systems and a satisfactory agreement between the calculated results and most of the experimental data was obtained.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at http://dx.doi.org/10.1016/j.calphad.2014.07.002.

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